



One point per cluster spatially balanced sampling

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ABSTRACT

A spatial sampling design determines where sample locations are placed in a study area so that population parameters can be estimated with relatively high precision. Spatially balanced designs have good spatial spread and give precise results for commonly used estimators when surveying natural resources. A new design is proposed which draws spatially balanced samples from stratified and unstratified populations. The method is two-fold. First, the population is partitioned into n compact geographic clusters. Then, a one point per cluster sample is drawn using a linear assignment strategy that optimises the spatial spread of the sample. Numerical results on several simulated populations show that the method generates well-spread samples and compares favourably with existing designs. An example application is also provided, where soil organic matter concentrations are estimated over a study area in Voorst, Netherlands.

1. Introduction

Spatial sampling designs determine where sample locations are placed in a study region so population characteristics can be estimated with relatively high precision. Usually, the population mean or total is estimated, but other characteristics may also be of interest. A straightforward approach is simple random sampling (SRS), where every possible subset of n locations has the same probability of being selected as the sample. Although SRS yields unbiased estimates of population characteristics, its precision can be relatively low, and there is no guarantee of a good geographical spreading. Many sampling designs have been proposed that improve SRS in various ways (cf. Thompson, 2012). An effective strategy for natural resources is a spatially balanced sampling (Stevens and Olsen, 2004), where locations are spread evenly over the resource. Spatial balance (SB) designs improve the precision of commonly used estimators for natural resources because nearby locations tend to have more similar response values than distant ones (Stevens and Olsen, 2004). This paper considers drawing SB samples from a finite set of locations.

SB designs are known to perform well when surveying natural resources (cf. Stevens and Olsen, 2004; Grafström and Lundström, 2013a; Grafström and Schelin, 2014) and a variety of designs have been proposed (cf. Benedetti et al., 2015; Kermorvant et al., 2019). One of the first and most popular SB designs is generalized random tessellation stratified (GRTS, Stevens and Olsen, 2004). GRTS is particularly useful for surveying natural resources because SB over-samples can be drawn if non-response is anticipated (Stevens and Olsen, 2004; Larsen et al., 2008). Several methods map the population to the real line in ways that preserve neighbourhood relationships and then sample the one-dimensional frame using, for example, systematic sampling (Stevens and Olsen, 1999, 2003, 2004; Theobald et al., 2007; Dickson and Tillé, 2016). Benedetti and Piersimoni (2017a) use a within-sample distance measure to spread samples evenly over a resource. Another group of methods apply a local repulsion strategy to discourage neighbouring units from appearing together in a sample (Grafström, 2011; Grafström et al., 2012; Grafström and Tillé, 2013b; Grafström and

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Matei, 2018). Designs using quasi-random number sequences (Robertson et al., 2013, 2017, 2021b; van Dam-Bates et al., 2018) and properties of these sequences (Robertson et al., 2018, 2021a) have also been proposed.

Another way to spread a sample over a population is to use spatial coverage sampling or spatial infill sampling (Brus, 2022). To draw a sample of n locations, the population is partitioned into n strata or clusters and one or two locations are selected from each cluster. A popular technique for partitioning the population is the k -means clustering algorithm (Brus et al., 2007; Walvoort et al., 2010; Benedetti et al., 2017b). Probability samples are not strictly required for mapping and model-based inference, so the most central unit from each cluster is usually selected to achieve optimal spatial coverage (Brus, 2022). However, for probability sampling and design-based inference, one unit is selected randomly from each cluster, and spatial coverage is often suboptimal (Benedetti et al., 2017b; Brus, 2022). This is a particular case of stratified SRS, with inclusion probabilities determined by cluster sizes. Hence, each cluster must have the same number of units to draw an equal probability sample. This can be achieved, for example, using a constrained k -means algorithm that respects specified cluster sizes (Walvoort et al., 2010; Benedetti et al., 2017b; Borgwardt et al., 2017; Brus, 2022).

We present a new sampling strategy for one point per cluster designs. Rather than randomly drawing one point from each cluster, our method randomly draws its sample from a collection of candidate samples. The advantage of this counterintuitive approach is that we can adapt the candidate samples without affecting inclusion probabilities. Specifically, we can exchange points (stochastically or deterministically) between the candidate samples to optimise a criterion for evaluating the collection. In this paper, we exchange points to improve spatial balance, although other objectives are possible within the given framework, including balancing on auxiliary totals or complex functions. Exchanging points between candidate samples to optimise an affine objective is a linear assignment problem, and optimal exchanges are tractable. The simulated average spatial balance of samples drawn using our linear assignment method is substantially better than what is achieved using random draws, and targeted inclusion probabilities are respected.

We begin by describing two popular ways of measuring spatial balance. The proposed one point per cluster design is presented in Section 3, with clustering methods in Section 3.1 and the linear assignment sampling strategy in Section 3.2. Numerical comparisons with several existing methods are given in Section 4, and the proposed method is implemented on a stratified population in Section 5, where soil organic matter concentrations are estimated over a study area in Voorst, Netherlands. Concluding remarks are given in Section 6.

2. Spatial balance

Let $\Omega = \{\mathbf{x}_1, \dots, \mathbf{x}_N\} \subset \mathbb{R}^2$ be a finite set of sampling locations, where $s \subset \Omega$ is a probability sample of n points and $0 < \pi_i < 1$ is the inclusion probability of $\mathbf{x}_i$ such that $\sum_i \pi_i = n$. In this paper, we consider two popular SB measures for s .

The first SB measure is based on a Voronoi tessellation of Ω using s (Stevens and Olsen, 2004). A sample is considered well-spread over Ω if the total inclusion probability of population points in each Voronoi polygon is approximately equal to one. Hence, each $\mathbf{x}_i \in s$ should have approximately N/n nearest neighbours in the population for equal probability sampling. The SB measure is defined by

$$m_1(s) = \frac{1}{n} \sum_{i=1}^n (v_i - 1)^2, \quad (1)$$

where v_i is the total inclusion probability of all points in the i th Voronoi polygon

$$v_i = \{\mathbf{x} \in \Omega : \|\mathbf{x} - \mathbf{x}_i\| \leq \|\mathbf{x} - \mathbf{x}_j\| \text{ for all } j = 1, \dots, n\}.$$

If $\mathbf{x} \in \Omega$ is in multiple Voronoi polygons, its inclusion probability is divided equally among the polygons that include it. No threshold value for (1) determines whether a sample is spatially balanced, but values closer to zero tend to indicate better spatial spread. Hence, this measure is helpful to compare spatial balances between different samples.

The second SB measure uses the normalised version of Moran's I Index proposed by Tillé et al. (2018) and is defined using a spatial weight matrix. If $\mathbf{x}_i$ were included in sample s , $\mathbf{x}_i$ would represent approximately $1/\pi_i$ points in Ω . Hence, for a well-spread sample, we would expect $\mathbf{x}_i$ to have approximately $h_i = \min\{1/\pi_i - 1, N - 1\}$ nearest neighbours in Ω . This heuristic is captured in a $N \times N$ spatial weight matrix

$$\mathbf{W} = \begin{pmatrix} 0 & w_{1,2} & \dots & w_{1,N} \\ w_{2,1} & 0 & \dots & w_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ w_{N,1} & w_{N,2} & \dots & 0 \end{pmatrix},$$

where spatial weights are calculated using

$$w_{i,j} = \begin{cases} 1 & \text{if } \mathbf{x}_j \text{ is in the set of } [h_i] \text{ nearest neighbours of } \mathbf{x}_i \text{ in } \Omega \setminus \mathbf{x}_i \\ h_i - [h_i] & \text{if } \mathbf{x}_j \text{ is the } ([h_i] + 1)\text{th nearest neighbour of } \mathbf{x}_i \text{ in } \Omega \setminus \mathbf{x}_i \\ 0 & \text{otherwise,} \end{cases}$$

and $[h]$ is the largest integer less than or equal to h . Hence, the i th row of $\mathbf{W}$ identifies $\mathbf{x}_i$'s neighbours and points with larger inclusion probabilities have fewer neighbours.

Let $\mathbf{a}$ be a $N \times 1$ sample indicator vector with

$$a_i = \begin{cases} 1 & \text{if } \mathbf{x}_i \in s \\ 0 & \text{otherwise,} \end{cases}$$

and let $\mathbf{1}$ be a $N \times 1$ vector of ones. The SB measure is defined by

$$m_2(s) = \frac{(\mathbf{a} - \bar{\mathbf{a}})^T \mathbf{W} (\mathbf{a} - \bar{\mathbf{a}})}{\sqrt{(\mathbf{a} - \bar{\mathbf{a}})^T \mathbf{D} (\mathbf{a} - \bar{\mathbf{a}}) (\mathbf{a} - \bar{\mathbf{a}})^T \mathbf{B} (\mathbf{a} - \bar{\mathbf{a}})}}, \quad (2)$$

where

$$\bar{\mathbf{a}} = \left(\frac{\mathbf{a}^T \mathbf{W} \mathbf{1}}{\mathbf{1}^T \mathbf{W} \mathbf{1}} \right) \mathbf{1}$$

is a weighted sample indicator mean, $\mathbf{D}$ is a diagonal matrix with $d_{i,i} = \sum_{j=1}^N w_{i,j}$,

$$\mathbf{B} = \mathbf{C}^T \mathbf{D} \mathbf{C} \quad \text{and} \quad \mathbf{C} = \mathbf{D}^{-1} \mathbf{W} - \frac{\mathbf{1} \mathbf{1}^T \mathbf{W}}{\mathbf{1}^T \mathbf{W} \mathbf{1}}.$$

This measure is on a standardised scale $-1 \leq m_2(s) \leq 1$ and interpretable, where negative values indicate spatial spread, positive values indicate spatial clustering and values close to zero are expected under SRS (Tillé et al., 2018). Hence, the closer $m_2(s)$ is to -1 , the greater the spatial spread of s over Ω .

In this paper, the proposed sampling design performs many calculations to optimise the spatial spread of candidate samples. Therefore, an efficient way to repeatedly compute (2) without requiring large matrix multiplications is desirable. Because $\mathbf{a}$ is a sparse indicator vector, an efficient way to compute (2) is

$$m_2(s) = \frac{\sum_{i,j \in S} w_{i,j} - \bar{a} \sum_{i \in S} z_i}{\sqrt{u \bar{a} (1 - \bar{a}) \left(\sum_{i,j \in S} b_{i,j} - 2 \bar{a} \sum_{i \in S} v_i + \bar{a}^2 b \right)}},$$

where $S = \{i : a_i = 1\}$ and $\bar{a} = \sum_{i \in S} d_{i,i} / u$. The vectors $\mathbf{z} = \mathbf{W}^T \mathbf{1}$, $\mathbf{v} = \mathbf{B}^T \mathbf{1}$ and the constants $u = \mathbf{1}^T \mathbf{W} \mathbf{1}$ and $b = \mathbf{1}^T \mathbf{B} \mathbf{1}$ only need to be computed once and stored. Hence, evaluating m_2 many times on the same population is not computationally prohibitive.

3. One point per cluster spatially balanced sampling

We present a new SB design that draws well-spread samples from stratified and unstratified populations. The method is two-fold. First, the population is grouped into n clusters, where n is the desired sample size. Then, one point is drawn from each cluster using a linear assignment strategy that optimises the spatial balance of the sample. The design cannot achieve arbitrary inclusion probabilities but can draw equal probability samples and unequal probability samples via stratification. We begin by describing clustering techniques, and the linear assignment sampling strategy is given in Section 3.2.

3.1. Spatial clustering

To implement our one point per cluster design, Ω is grouped into n clusters. We consider clusterings of the form

$$\{C_1, \dots, C_n\} \subset \Omega \text{ with } C_i \cap C_j = \emptyset \text{ for } i \neq j, \quad (3)$$

where the clustering can be done on the full population or a subset of Ω . There are computational advantages to clustering a subset of Ω , but it also allows us to draw equal probability samples. We cluster an SRS of λn points from Ω into n clusters containing λ points each, where λ is an integer satisfying $\lambda n \leq N$. For relatively small Ω , we choose λ as large as possible (see Fig. 1). The inclusion probabilities for a one point per cluster design are $(\lambda n / N) / \lambda = n / N$ as required.

There are many possible ways to cluster a set of locations. This paper focuses on drawing SB samples, so defining clusters as collections of nearby points is a natural choice. However, auxiliary information like slope, temperature, elevation, etc. can also be incorporated into the spatial clustering (cf. Brus, 2022). We use the popular k -means algorithm to define compact geographical clusters.

The k -means algorithm partitions Ω into n clusters by iteratively updating cluster assignments to reduce within-cluster variation. The number of points N_k in the k th cluster C_k is determined implicitly by the algorithm and depends heavily on the spatial structure of the population (see Fig. 1). Hence, a one point per k -means cluster design will be an unequal probability design, where $\pi_k = 1 / N_k$ for all $\mathbf{x} \in C_k$. To control these inclusion probabilities, a constrained k -means algorithm is used (Walvoort et al., 2010; Benedetti et al., 2017b; Borgwardt et al., 2017; Brus, 2022). This can be expressed as an optimisation problem of the form

$$\begin{aligned} & \text{minimize} && \sum_{k=1}^n \sum_{i=1}^N z_{ki} \|\mathbf{x}_i - \mathbf{m}_k\|^2 \\ & \text{subject to} && \sum_{i=1}^N z_{ki} = N_k, \\ & && \sum_{k=1}^n z_{ki} = 1, \\ & && z_{ki} \in \{0, 1\} \forall i, k, \end{aligned} \quad (4)$$

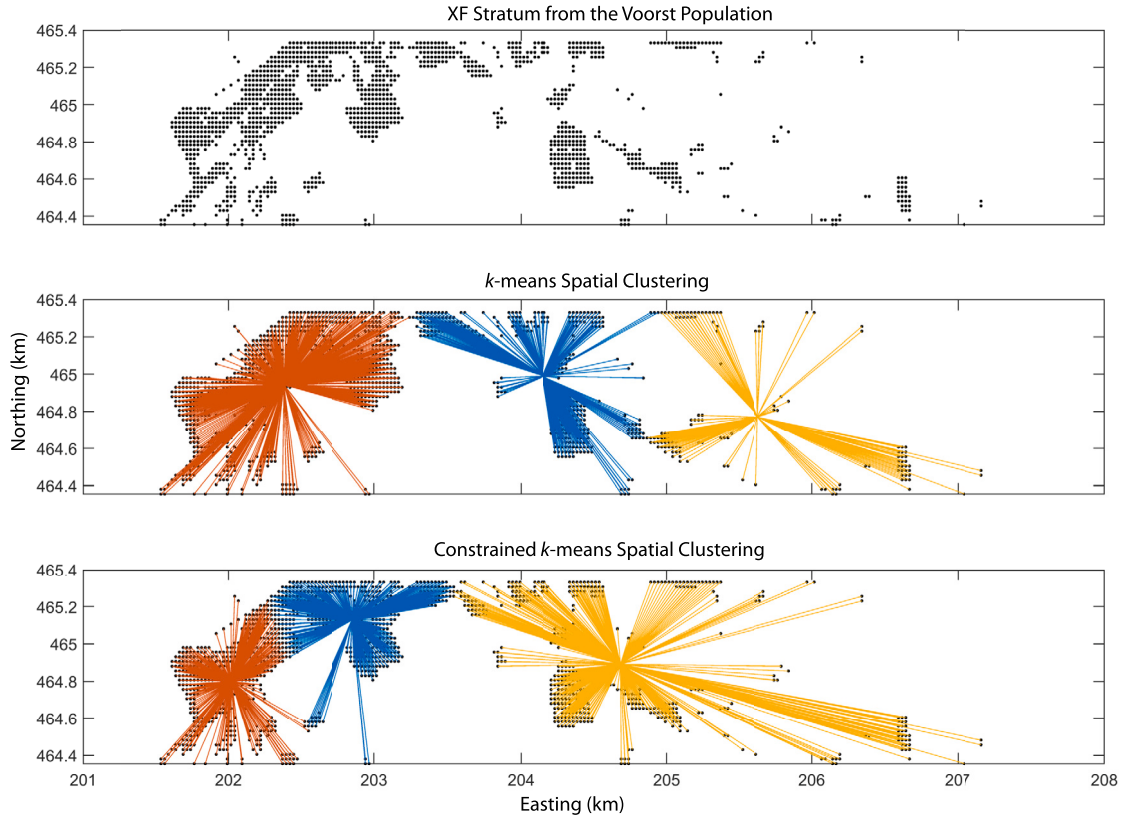


Fig. 1. k -means clustering applied to the XF stratum from the Voorst population (see Section 5), where $N = 1346$ locations are grouped into $n = 3$ compact geographical clusters. k -means has wildly different cluster sizes, where $N_1 = 846$ (orange), $N_2 = 365$ (blue) and $N_3 = 135$ (yellow). Constrained k -means has $N_k = 448$ in each cluster, where two points were randomly removed to ensure equal inclusion probabilities for a one point per cluster design. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

where $\mathbf{m}_k$ is the mean of the k th cluster. The first constraint controls inclusion probabilities by forcing N_k points in the k th cluster, and the second ensures each point is assigned to a cluster (see Fig. 1). Problem (4) can be solved efficiently by alternating between the following two steps until the cluster assignments do not change or the within-cluster variation is sufficiently small.

1. For a given cluster assignment (initially random but satisfying the constraints), compute the mean of each cluster.
2. Assign each point to its closest (current) cluster mean while satisfying the constraints.

The only difference between this algorithm and the standard k -means algorithm is that the assignment step (step 2) needs to satisfy the first constraint. This assignment step can be solved using linear programming methods (Borgwardt et al., 2017), a modified k -means algorithm (Walvoort et al., 2010; Benedetti et al., 2017b) or more efficiently by expressing (4) as a linear assignment problem and using the Hungarian algorithm (Malinen and Fränti, 2014) or its variants (Duff and Koster, 2001).

The k -means algorithm is a stochastic, local optimisation method that is sensitive to initial cluster assignments. Hence, it is common practice to run the algorithm several times with different initial conditions and take the best clustering found. However, we only require rough clusters of nearby points, which can be achieved with a few iterations of k -means from any initial cluster assignments that satisfy the constraints. If initial cluster means are provided in targeted regions of the study area, only one assignment step is required.

For stratified populations, the clustering method above is applied independently for each stratum (see Fig. 1), where the number of clusters is set equal to the stratum's sample size. Clustering each stratum separately allows our one point per cluster design to draw stratified random samples, with equal probability samples drawn from each stratum. An advantage of our design is that we can direct the sample to be simultaneously well-spread over Ω and within each stratum. Stratified populations are investigated in Section 5.

3.2. Drawing one point per cluster samples

Existing cluster-based designs usually select one point randomly from each cluster, and spatial coverage is often suboptimal (Benedetti et al., 2017b; Brus, 2022). Our approach is markedly different because it draws its sample from a collection of J candidate

samples optimised for SB. For most applications, $J = \max(N_k)$, but having the flexibility to set $J < \max(N_k)$ improves computational performance on large populations (see Section 3.2.1). The candidate samples are of the form

$$S = \begin{cases} s^{(1)} &= \{ \mathbf{x}_1^{(1)}, \mathbf{x}_2^{(1)}, \dots, \mathbf{x}_n^{(1)} \} \\ s^{(2)} &= \{ \mathbf{x}_1^{(2)}, \mathbf{x}_2^{(2)}, \dots, \mathbf{x}_n^{(2)} \} \\ \vdots &\vdots \\ s^{(J)} &= \{ \mathbf{x}_1^{(J)}, \mathbf{x}_2^{(J)}, \dots, \mathbf{x}_n^{(J)} \}, \end{cases} \quad (5)$$

where superscripts and subscripts indicate candidate sample and cluster memberships, respectively. For example, $\mathbf{x}_k^{(j)}$ is in the j th candidate sample $s^{(j)}$ and from the k th cluster C_k . Hence, each $s^{(j)}$ is a one point per cluster sample from Ω . The points in S from the k th cluster

$$p_k = \{ \mathbf{x}_k^{(1)}, \mathbf{x}_k^{(2)}, \dots, \mathbf{x}_k^{(J)} \},$$

are randomly drawn from C_k . This is done by placing the points in k th cluster in random order, $C_k = \{ \mathbf{x}_k^{(1)}, \dots, \mathbf{x}_k^{(N_k)} \}$, and selecting the first J points from C_k ; returning to the beginning of the sequence if $J > N_k$. If $J \leq N_k$, p_k is an SRS from C_k . Otherwise, repetitions occur in p_k with each $\mathbf{x} \in C_k$ included at least $\lambda_k = \lfloor J/N_k \rfloor$ times, but no more than $\lambda_k + 1$ times. Choosing $J = \max(N_k)$ ensures each point in the clustering is included in S .

Randomly choosing one candidate sample from S gives equal inclusion probabilities for each $\mathbf{x} \in C_k$ of the form

$$\pi_k = \beta_k \frac{\lambda_k}{J} + (1 - \beta_k) \frac{\lambda_k + 1}{J} = \frac{1}{N_k}, \quad (6)$$

where $\beta_k = \lambda_k + 1 - J/N_k$ is the probability that $\mathbf{x}$ is included in λ_k candidate samples. However, (6) also holds if we permute (stochastically or deterministically) each p_k . Hence, we can exchange points between the candidate samples using

$$p_k = \{ \mathbf{x}_k^{(\sigma_1)}, \mathbf{x}_k^{(\sigma_2)}, \dots, \mathbf{x}_k^{(\sigma_J)} \}, \quad (7)$$

where σ is a permutation of $\{1, \dots, J\}$ and the j th candidate sample is replaced with

$$s^{(j)} = \{ \mathbf{x}_1^{(j)}, \dots, \mathbf{x}_k^{(\sigma_j)}, \dots, \mathbf{x}_n^{(j)} \}. \quad (8)$$

The next section provides an optimal permutation that improves spatial balance.

3.2.1. Linear assignments

Assignment problems determine how to assign J items to J objects in the best possible way. Every permutation $\sigma : \{1, \dots, J\} \rightarrow \{1, \dots, J\}$ is a possible assignment, represented by

$$\begin{pmatrix} 1 & 2 & \dots & J \\ \sigma_1 & \sigma_2 & \dots & \sigma_J \end{pmatrix},$$

where item j is assigned to object σ_j . In this paper, optimal permutations of the form (7) are proposed for each p_k to minimise the average SB of S . Rather than minimising (1) or (2) directly, we consider a convex combination of these measures, given by

$$m(s) = \alpha m_1(s) + (1 - \alpha) m_2(s), \quad (9)$$

where $0 \leq \alpha \leq 1$. Setting $\alpha = 1$ minimises Voronoi SB, $\alpha = 0$ minimises modified Moran's I and choosing $0 < \alpha < 1$ incorporates aspects from both SB measures into the sample.

Consider assigning J points from the k th cluster to J candidate samples, where the k subscripts on $\mathbf{x}$ are omitted for ease of exposition. The cost of assigning $\mathbf{x}^{(\sigma_j)}$ to candidate sample $s^{(j)}$ using (8) is calculated using (9) and denoted $m(s^{(j)}, \mathbf{x}^{(\sigma_j)})$. The cost matrix for all updates of this form on the k th cluster is

$$\mathbf{C} = \begin{pmatrix} m(s^{(1)}) & m(s^{(1)}, \mathbf{x}^{(2)}) & \dots & m(s^{(1)}, \mathbf{x}^{(J)}) \\ m(s^{(2)}, \mathbf{x}^{(1)}) & m(s^{(2)}) & \dots & m(s^{(2)}, \mathbf{x}^{(J)}) \\ \vdots & \vdots & \ddots & \vdots \\ m(s^{(J)}, \mathbf{x}^{(1)}) & m(s^{(J)}, \mathbf{x}^{(2)}) & \dots & m(s^{(J)}) \end{pmatrix}, \quad (10)$$

where the initial SBs for each $s^{(j)}$ (no update, $\sigma_j = j$) are down the main diagonal.

An optimal permutation for (7) is one that minimises the average SB of S . This is found using a linear integer programme of the form

$$\begin{aligned} &\text{minimize} && \sum_{j=1}^J \sum_{i=1}^J z_{ji} m(s^{(j)}, \mathbf{x}^{(i)}) \\ &\text{subject to} && \sum_{j=1}^J z_{ji} = 1, \\ &&& \sum_{i=1}^J z_{ji} = 1, \\ &&& z_{ji} \in \{0, 1\} \forall i, j, \end{aligned} \quad (11)$$

where the constraints ensure each $\mathbf{x}^{(j)}$ is uniquely assigned to one candidate sample. Solving (11) is equivalent to permuting the rows or columns of $\mathbf{C}$ to minimise its trace. Let $\mathbf{e}_j$ be the j th canonical basis vector. If $\mathbf{P} = (\mathbf{e}_{\sigma_1}, \dots, \mathbf{e}_{\sigma_J})$ is a permutation matrix that minimises the trace of $\mathbf{PC}$, then each candidate sample in (5) is updated using (8).

Example 3.1. Consider assigning four points, $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(4)}$, to four candidate samples, $s^{(1)}, \dots, s^{(4)}$. Assume the cost matrix for assigning each point to each candidate sample is

$$\mathbf{C} = \begin{matrix} & \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \mathbf{x}^{(3)} & \mathbf{x}^{(4)} \\ \begin{pmatrix} 0.0 & 0.3 & -0.1 & 0.1 \\ -0.2 & 0.1 & -0.3 & 0.2 \\ 0.3 & -0.2 & -0.4 & 0.5 \\ -0.1 & -0.1 & 0.3 & -0.1 \end{pmatrix} & \begin{matrix} s^{(1)} \\ s^{(2)} \\ s^{(3)} \\ s^{(4)} \end{matrix} \end{matrix},$$

where, for example, the cost of assigning $\mathbf{x}^{(4)}$ to $s^{(3)}$ is 0.5. The permutation matrix $\mathbf{P} = (\mathbf{e}_4, \mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2)$ minimises the trace of the permuted cost matrix, where

$$\mathbf{PC} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0.0 & 0.3 & -0.1 & 0.1 \\ -0.2 & 0.1 & -0.3 & 0.2 \\ 0.3 & -0.2 & -0.4 & 0.5 \\ -0.1 & -0.1 & 0.3 & -0.1 \end{pmatrix} = \begin{pmatrix} -0.2 & 0.1 & -0.3 & 0.2 \\ -0.1 & -0.1 & 0.3 & -0.1 \\ 0.3 & -0.2 & -0.4 & 0.5 \\ 0.0 & 0.3 & -0.1 & 0.1 \end{pmatrix} \begin{matrix} s^{(2)} \\ s^{(4)} \\ s^{(3)} \\ s^{(1)} \end{matrix}.$$

Hence, $\sigma = (4, 1, 3, 2)$ gives optimal assignments with $\mathbf{x}^{(4)} \in s^{(1)}$, $\mathbf{x}^{(1)} \in s^{(2)}$, $\mathbf{x}^{(3)} \in s^{(3)}$ and $\mathbf{x}^{(2)} \in s^{(4)}$. The initial average SB of -0.1 was reduced to -0.15 following the update.

Assignment problems can be solved efficiently using the Hungarian algorithm and its variants (Duff and Koster, 2001), provided J is not too large (worst-case complexity $\mathcal{O}(J^3)$). To indicate the time required to solve assignment problems, we report the average times needed to generate the results in Section 4 using Matlab's matchpairs solver. For $n = 20$ ($J = 50$), the algorithm took approximately 0.04 seconds to solve all 60 assignment problems needed to draw a sample. For $n = 200$ ($J = 5$), 600 assignment problems were solved, taking approximately 0.06 seconds. Although large assignment problems are tractable (roughly 2 seconds for $J = 1000$), there are diminishing returns for the additional computational effort because nearby points have similar assignment costs. Hence, many assignments give marginal improvements to the average sample spread in S for large J . For large populations, $J \leq 500$ should be sufficient.

3.3. Sampling design

The sampling design is given in Algorithm 1. At line 3, J candidate samples, each containing one point per cluster, are drawn from Ω using (5). The inner loop (lines 7 to 12) solves n linear assignment problems (one for each cluster), and line 6 randomises the order in which these assignment problems are solved to increase sample entropy. The outer loop (lines 5 to 13) executes M times, giving a total of nM assignment problems. For each problem, we minimise the average SB of S , $\bar{m} = J^{-1} \sum_{j=1}^J m(s^{(j)})$, by solving (11). This generates a sequence $\bar{m}_0, \dots, \bar{m}_{nM}$ of non-increasing SB averages, but there is no guarantee that $\bar{m}_{nM}$ is a minimum in the classical sense because each minimisation is conditional on the remaining $n - 1$ cluster assignments being fixed. However, accepting suboptimal solutions with relatively small M values, rather than seeking a local or global minimum, reduces computational effort and increases sample entropy. We only require a sufficient reduction in $\bar{m}$ to draw SB samples, and relatively small $M \leq 3$ values achieve this (see Section 4.1).

After M iterations of the outer loop, the candidate samples in S are optimised for SB. A sample is obtained by randomly choosing one (or more) from S .

3.4. Estimation

The Horvitz-Thompson estimator

$$\hat{\tau} = \sum_{k=1}^n \frac{y_k}{\pi_k} \quad (12)$$

is an unbiased estimator of the population total τ , where y_k is the response value for $\mathbf{x}_k \in \Omega$. The variance of (12) can be estimated for SB designs, but an unbiased estimator is not always possible because second-order inclusion probabilities for nearby points can be zero (Grafström and Schelin, 2014; Robertson et al., 2018). We provide two commonly used SB design estimators and a resampling estimator. Our design produces SB samples with similar spatial properties to existing designs and has known first-order inclusion probabilities, so these estimators are also suitable here.

Algorithm 1 One point per cluster spatially balanced sampling.

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1: Input: A clustering  $\{C_1, \dots, C_n\} \subset \Omega$  using (3), with  $C_k$  containing  $N_k$  points.
2: Choose the number of candidate samples  $2 \leq J \leq \max(N_k)$  and iterations  $M \geq 1$ .
3: Define  $J$  candidate one point per cluster samples  $S = \{s^{(1)}, \dots, s^{(J)}\}$  using (5), where each candidate sample is of the form  $s^{(j)} = \{\mathbf{x}_1^{(j)}, \dots, \mathbf{x}_n^{(j)}\}$  with  $\mathbf{x}_k^{(j)} \in C_k$ .
4: Set  $r = 0$  and compute the initial average SB  $\bar{m}_0 = J^{-1} \sum_{j=1}^J m(s^{(j)})$ .
5: for  $i = 1$  to  $M$  do
6:   Let  $\beta$  be a random permutation of  $\{1, \dots, n\}$ .
7:   for  $k \in \beta$  do
8:     Select the  $J$  points in  $S$  from the  $k$ th cluster,  $p_k = \{\mathbf{x}_k^{(1)}, \dots, \mathbf{x}_k^{(J)}\}$ .
9:     Solve assignment problem (11) to find a permutation  $\sigma$  of  $\{1, \dots, J\}$  for  $p_k$ .
10:    Update each  $s^{(j)} \in S$  using  $s^{(j)} = \{\mathbf{x}_1^{(j)}, \dots, \mathbf{x}_k^{(\sigma_j)}, \dots, \mathbf{x}_n^{(j)}\}$ .
11:    Set  $r = r + 1$  and compute average SB  $\bar{m}_r$ .
12:   end for
13: end for
14: Output: Sample  $s^{(j)} \in S$ , where  $j \in \{1, \dots, J\}$  is randomly chosen.

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The local mean variance estimator (Stevens and Olsen, 2003) is

$$\widehat{\text{var}}_{\text{NBH}}(\hat{\tau}) = \sum_{k=1}^n \sum_{i \in D_k} w_{ki} \left(\frac{y_i}{\pi_i} - \hat{\tau}_{D_k} \right)^2,$$

where D_k is a neighbourhood of $\mathbf{x}_k$ containing at least four points from the sample, $\hat{\tau}_{D_k}$ is an estimated total using the response values from the points in D_k and w_{ki} are spatial weights that decrease as the distance between $\mathbf{x}_k$ and $\mathbf{x}_i$ increases. The second estimator is based on squared local deviations (Grafström and Schelin, 2014), given by

$$\widehat{\text{var}}_{\text{SB}}(\hat{\tau}) = \sum_{k=1}^n \frac{n_k^*}{n_k^* - 1} \left(\frac{y_k}{\pi_k} - \frac{1}{n_k^*} \sum_{i \in s_k^*} \frac{y_i}{\pi_i} \right)^2,$$

where s_k^* is a subset of the sample containing $n_k^* \geq 2$ points. This subset includes all the sample points satisfying $\|\mathbf{x} - \mathbf{x}_k\| \leq d$ (including the k th point itself), where d is the distance to the nearest neighbour in the sample.

The final estimator is a resampling method proposed by Robertson et al. (2021b, 2022). The method draws B resamples from the population using the design that selected the original sample. Then, each resampled point is assigned the response value of its nearest neighbour (Euclidean distance) in the original sample. For each resample, a Horvitz-Thompson estimate $\hat{\tau}_b^*$ is calculated and the variance estimator is

$$\widehat{\text{var}}_B = \frac{1}{B} \sum_{b=1}^B (\hat{\tau}_b^* - \bar{\tau}^*)^2,$$

where $\bar{\tau}^* = \frac{1}{B} \sum_{b=1}^B \hat{\tau}_b^*$. Results in Section 4.3 were generated using $B = 200$.

4. Numerical results

In this section, we investigate the statistical properties of the proposed one point per k -means cluster design. We used two typical spatial structures for $N = 1000$ points over the unit box (see Fig. 2).

1. Regular population: A BAS sample (Robertson et al., 2013) of $N = 1000$ points from $[0, 1]^2$.
2. Aggregated population: $N = 1000$ points generated over $[0, 1]^2$ according to a conditional Neyman-Scott point process (Neyman and Scott, 1958) with 100 clusters and 10 points per cluster.

4.1. Rate of convergence

The parameter M in Algorithm 1 determines how many linear assignment problems are solved to improve the average SB of S . This parameter directly affects the computational performance of the sampling design, and unnecessarily large values should be avoided. To investigate the sequence $\{\bar{m}_r\}_{r=1}^M$ generated by Algorithm 1, we selected equal probability samples from the regular and aggregated populations. Results using $\alpha = 0$ in (9) (minimising m_2) are presented in Table 1.

The average SB for $M = 0$ corresponds to a design that randomly chooses one point from each k -means cluster. Hence, the k -means clustering performed well on these populations. However, the spatial balance was substantially improved using the linear assignment technique, with significant gains made after just one iteration (n linear assignments). After $M = 3$ iterations, minimal gains were made. Similar results were observed for other α values, where three iterations yielded a sufficient reduction in (9). Hereafter all results were generated using $M = 3$ and $J = \max(N_k)$ in (10). Therefore, $3n$ linear assignment problems were solved to generate each batch of J candidate samples.

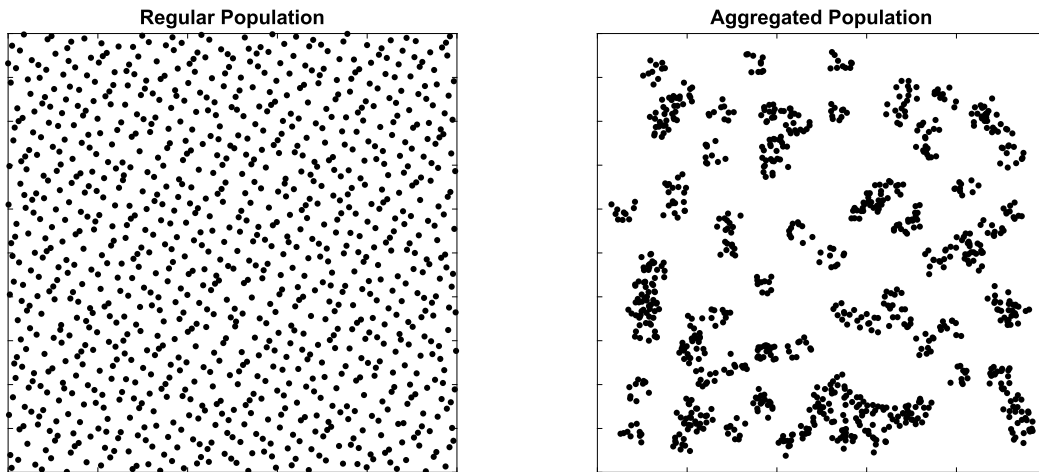


Fig. 2. Populations of $N = 1000$ points used in the simulation study.

Table 1

Rate of convergence of the sequence $\{\bar{m}_r\}_{r=1}^M$ generated by Algorithm 1 with $\alpha = 0$, where $M = 0$ is the initial average spatial balance $\bar{m}_0$. All values are averages over 10,000 samples.

n	M						
	0	1	2	3	4	5	6
<i>Regular Population</i>							
20	-0.13	-0.31	-0.34	-0.35	-0.35	-0.35	-0.35
50	-0.12	-0.43	-0.46	-0.47	-0.48	-0.48	-0.48
100	-0.29	-0.55	-0.58	-0.59	-0.60	-0.60	-0.60
200	-0.37	-0.67	-0.70	-0.71	-0.72	-0.72	-0.72
<i>Aggregated Population</i>							
20	-0.15	-0.36	-0.40	-0.41	-0.42	-0.42	-0.42
50	-0.23	-0.54	-0.58	-0.60	-0.60	-0.60	-0.60
100	-0.39	-0.66	-0.68	-0.69	-0.69	-0.69	-0.69
200	-0.41	-0.71	-0.73	-0.73	-0.73	-0.73	-0.73

4.2. Spatial balance

To compare the spatial balance of the proposed design with existing methods, we selected equal probability samples from the regular and aggregated populations (see Fig. 2) using four sampling designs.

1. Generalized Random Tessellation Stratified (GRTS) using the *R* package *spsurvey* (Dumelle et al., 2022).
2. Local Pivotal Method (LPM) using the *R* package *BalancedSampling* (Grafström et al., 2022).
3. Halton Iterative Partitioning (HIP).
4. The proposed one point per k -means cluster (KMC) design.

For the KMC design, samples can be selected in several ways. We consider four ways: randomly selecting one point from each k -means cluster (Random Draw) and linear assignments with $\alpha = 0, 0.7$, and 1 in (9). Setting $\alpha = 1$ produces candidate samples that minimise Voronoi SB, and setting $\alpha = 0$ minimises modified Moran's I . The relationship between α and SB is illustrated in Fig. 3. The cloud of SB values for KMC samples was nicely separated from those generated by GRTS, HIP and LPM, with many KMC samples performing better on both measures. For the regular and aggregated populations, setting $\alpha = 0.7$ reduced the m_1 average without increasing the m_2 average too much. This met our objective of achieving a good spread on both measures and was used in the following simulations. Finding a suitable α requires testing several values and selecting the best one. Because Voronoi SB is not on a standardised scale, we present a relative measure by dividing the m_1 average of a design by the m_1 average under SRS. Results are presented in Tables 2 and 3.

All the designs had m_1 averages less than one and negative m_2 averages, indicating better average SB than SRS. Of the three existing designs (Table 2), LPM had the best SB averages on both populations. The random draw KMC design (Table 3) had similar m_1 averages to LPM, but LPM had better m_2 averages. The best designs for m_1 and m_2 were KMC designs using linear assignments with $\alpha = 1$ and $\alpha = 0$, respectively. These SB averages act as bounds on what is possible for the linear assignment strategy. The convex combination with $\alpha = 0.7$ had slightly worse m_2 averages than $\alpha = 0$, but much better m_1 averages than $\alpha = 0$. However, a

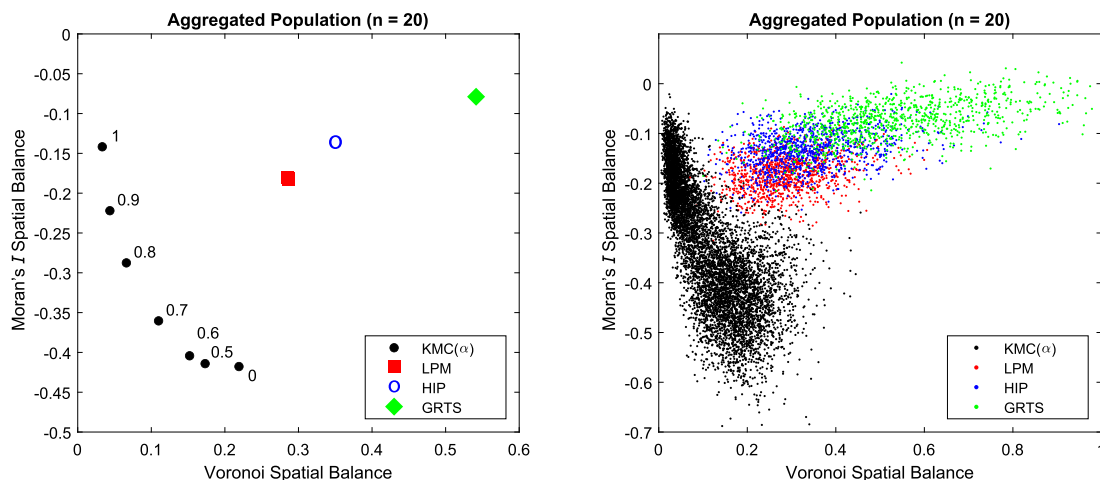


Fig. 3. Spatial balance of $n = 20$ points drawn from the aggregated population using GRTS, HIP, LPM and KMC designs with linear assignments and different α values in (9). **Left:** The average SB of each design is over 1000 samples. **Right:** The SB of each sample drawn by each design.

Table 2

Spatial balance of existing designs on the regular and aggregated populations. The method that performed best on m_1 and m_2 is shown in bold. All values are averages over 10,000 samples.

n	GRTS		HIP		LPM	
	m_1	m_2	m_1	m_2	m_1	m_2
<i>Regular Population</i>						
20	0.50	-0.08	0.25	-0.13	0.24	-0.16
50	0.48	-0.12	0.25	-0.20	0.23	-0.24
100	0.49	-0.15	0.30	-0.25	0.24	-0.32
200	0.56	-0.19	0.41	-0.29	0.31	-0.43
<i>Aggregated Population</i>						
20	0.54	-0.08	0.35	-0.13	0.28	-0.18
50	0.58	-0.10	0.41	-0.18	0.27	-0.28
100	0.60	-0.11	0.43	-0.22	0.21	-0.41
200	0.67	-0.10	0.48	-0.29	0.30	-0.46

Table 3

Spatial balance on the regular and aggregated populations for KMC designs using four sampling methods: random draw and linear assignments with $\alpha = 0, 0.7$, and 1 in (9). The method that performed best on m_1 and m_2 is shown in bold. All values are averages over 10,000 samples.

n	Random Draw		Linear Assignment					
	m_1	m_2	$\alpha = 1$ (Voronoi)		$\alpha = 0.7$		$\alpha = 0$ (Moran's I)	
			m_1	m_2	m_1	m_2	m_1	m_2
Regular Population								
20	0.23	-0.13	0.01	-0.14	0.05	-0.32	0.12	-0.35
50	0.23	-0.19	0.02	-0.22	0.06	-0.45	0.12	-0.47
100	0.26	-0.25	0.03	-0.31	0.08	-0.57	0.16	-0.59
200	0.34	-0.33	0.04	-0.39	0.13	-0.69	0.26	-0.71
Aggregated Population								
20	0.25	-0.16	0.03	-0.14	0.09	-0.36	0.22	-0.41
50	0.27	-0.26	0.07	-0.23	0.17	-0.57	0.24	-0.60
100	0.15	-0.40	0.03	-0.36	0.11	-0.65	0.17	-0.69
200	0.30	-0.42	0.09	-0.41	0.17	-0.67	0.30	-0.73

Table 4

Response total estimation on the regular and aggregated populations with $n = 50$. The method that performed best is shown in bold. All values are $\times 10^4$, and exact values are given for SRS. All other results are averages over 10,000 samples.

Function	SRS	GRTS		HIP		LPM	
	var	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{NBH}}$	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{SB}}$	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{SB}}$
<i>Regular Population</i>							
Plane	4.00	0.58	0.30	0.23	0.26	0.25	0.29
Peak	7.02	3.23	2.10	1.28	2.21	1.29	2.41
Bird	7.14	3.02	2.22	2.10	2.46	1.59	2.67
<i>Aggregated Population</i>							
Plane	3.90	0.37	0.20	0.21	0.16	0.17	0.18
Peak	7.76	2.26	1.67	1.43	1.65	0.86	1.89
Bird	7.20	2.56	2.39	1.86	2.21	1.02	2.52

Table 5

Response total estimation on the regular and aggregated populations for KMC designs using four sampling methods: random draw and linear assignments with $\alpha = 0, 0.7$, and 1 in (9). The method that performed best is shown in bold. All values are $\times 10^4$ and averages over 10,000 samples.

Function	Random Draw		Linear Assignment					
			$\alpha = 1$ (Voronoi)		$\alpha = 0.7$		$\alpha = 0$ (Moran's I)	
	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{B}}$	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{B}}$	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{B}}$	$\widehat{\text{var}}_{\text{sim}}$	$\widehat{\text{var}}_{\text{B}}$
<i>Regular Population</i>								
Plane	0.18	0.32	0.06	0.22	0.09	0.24	0.15	0.30
Peak	1.56	2.33	0.57	1.60	0.21	1.26	0.33	1.35
Bird	1.66	2.09	1.03	1.57	0.56	1.22	0.65	1.33
<i>Aggregated Population</i>								
Plane	0.10	0.15	0.08	0.15	0.10	0.15	0.11	0.15
Peak	0.94	1.11	0.64	0.87	0.37	0.69	0.44	0.77
Bird	1.18	1.43	0.99	1.06	0.56	0.97	0.59	1.02

computational cost is associated with convex combinations because m_1 and m_2 need to be computed for each linear assignment. The KMC design with $\alpha = 0$ is attractive because it is computationally efficient and produced m_1 and m_2 averages less than those achieved using existing methods.

4.3. Precision

To compare the precision of the proposed design with existing methods, we estimated population totals on the regular and aggregated populations (see Fig. 2) using GRTS, HIP, LPM and the KMC designs. Response values were defined via a continuous function $f : [0, 1]^2 \rightarrow \mathbb{R}$, where the i th response is $y_i = f(\mathbf{x}_i)$. Three functions from Robertson et al. (2013, 2018, 2022) were used (Plane, Peak and Bird), giving six populations with the Plane function having the strongest spatial trend. These functions are given and illustrated in the supplementary material.

To investigate precision, $\text{var}(\hat{\tau})$ was estimated for $n = 50$ using the empirical variance estimator over $R = 10,000$ samples

$$\widehat{\text{var}}_{\text{sim}}(\hat{\tau}) = \frac{1}{R} \sum_{i=1}^R (\hat{\tau}_i - \tau)^2, \quad (13)$$

where $\hat{\tau}_i$ is the estimated total for the i th sample. We also test the variance estimators from Section 3.4. Results are given in Tables 4 and 5.

GRTS, HIP and LPM offered substantial improvements over SRS for these populations, with LPM performing the best on average. The KMC designs also performed well. The simple one point per cluster KMC design using random draws achieved similar results to LPM, but linear assignments obtained better results. The convex combination with $\alpha = 0.7$ had slightly better $\widehat{\text{var}}_{\text{sim}}$ averages than $\alpha = 0$. However, a computational cost is associated with convex combinations because m_1 and m_2 must be computed for each linear assignment. The KMC design with $\alpha = 0$ was attractive because it is simpler and produced empirical variance averages less than those achieved using existing methods. The resampling estimator $\widehat{\text{var}}_{\text{B}}$ gave the best results for the KMC designs, with $\widehat{\text{var}}_{\text{NBH}}$ and $\widehat{\text{var}}_{\text{SB}}$ giving overly conservative estimates. Although this estimator had a positive bias, its average values were less than those obtained for existing designs and substantially less than an SRS estimator.

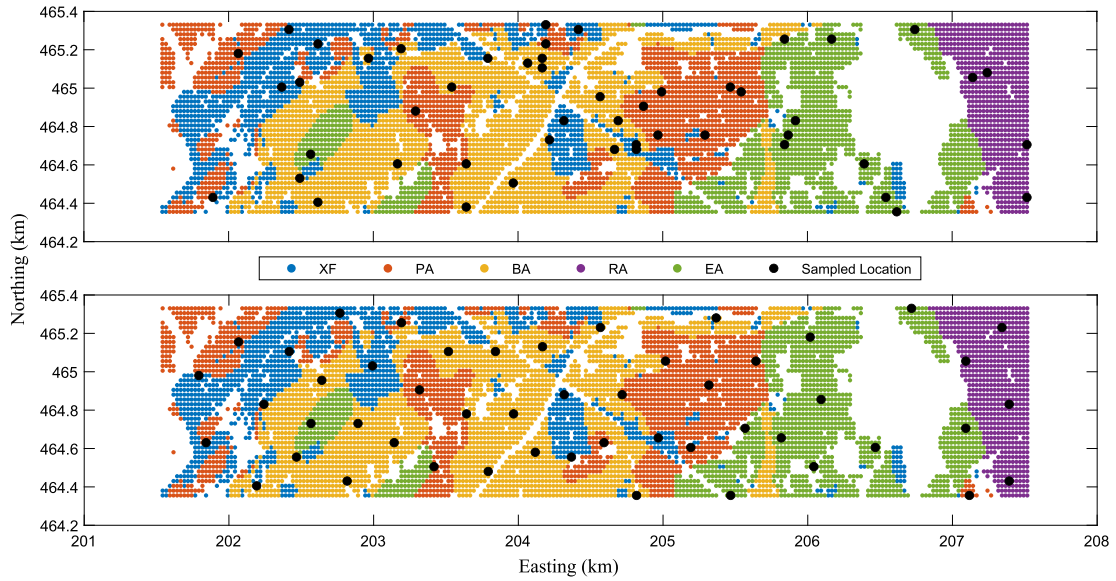


Fig. 4. Stratified Voorst study region with five soil-land use strata. **Top:** A stratified simple random sample of $n = 50$ locations using proportional allocation. **Bottom:** A one point per k -means cluster sample (proportional allocation) using linear assignments. The KMC design forces spatial spread and achieves desired stratum sample sizes.

5. Stratified Voorst population

To investigate the relationship between SB and precision for a stratified population, we consider the Voorst population (Brus, 2022). These data contain the spatial locations of $N = 7528$ points over a $6 \text{ km} \times 1 \text{ km}$ study area in Voorst, Netherlands (see Fig. 4). Auxiliary variables for soil class and land use are available at each location, which are combined to define five soil-land use strata: PA, BA, EA, RA, XF (Brus, 2022). The first letter of each stratum label denotes soil type:

- P for podzols (sandy soil with eluviated horizon below topsoil)
- B for beekkerdgrond (sandy wetland soil with gleyic properties)
- E for enkeerdgrond (sandy soil with thick anthropogenic humic topsoil)
- R for river clay soil
- X for other sandy soils

The second letter denotes land use: A for agriculture and F for forest. The response variable is soil organic matter (SOM) concentrations (in grams per kg of dry soil).

5.1. Spatial balance

We consider two measures of SB in this section. The first is the overall spatial balance of the sample using the modified Moran's I measure m_2 . The second is a measure of the average within soil-land use stratum SB, given by

$$\bar{m}_h(s) = \sum_{h=1}^5 \frac{n_h}{n} m_2(s^{(h)}), \quad (14)$$

where $s = \{s^{(1)}, \dots, s^{(5)}\}$, $s^{(h)}$ is the sub-sample of n_h locations from stratum h and $m_2(s^{(h)})$ is the modified Moran's I measure of sub-sample $s^{(h)}$ on stratum h .

We selected stratified random samples with proportional allocation to investigate spatial balance for the Voorst population (see Fig. 4). Inclusion probabilities within each stratum were equal but differed slightly between strata to achieve integer stratum sample sizes such that $\sum_h n_h = n$. We compared stratified SRS with three sampling designs.

1. Stratified GRTS using the *R* package *spsurvey* (Dumelle et al., 2022).
2. Stratified doubly balanced sampling using *lcubestratified* in the *R* package *BalancedSampling* (Grafström et al., 2022).
3. The proposed one point per k -means cluster (KMC) design.

Two ways of drawing KMC samples with linear assignments were considered. The first used (9) with $\alpha = 0$ (minimising m_2), and the second applied KMC independently within each soil-land use stratum (equivalent to replacing objective function (9) with (14) in the

Table 6

Spatial balance of stratified random sampling approaches with proportional allocation on the Voorst population. The method that performed best on m_2 and $\bar{m}_h$ is shown in bold. All values are averages over 10,000 samples.

n	Stratified SRS		Stratified GRTS		lcubestratified	
	m_2	$\bar{m}_h$	m_2	$\bar{m}_h$	m_2	$\bar{m}_h$
20	-0.01	-0.02	-0.02	-0.04	-0.03	-0.08
50	-0.01	-0.01	-0.03	-0.05	-0.06	-0.10
100	-0.01	-0.01	-0.05	-0.06	-0.09	-0.13
200	-0.00	-0.01	-0.07	-0.08	-0.14	-0.18
KMC Designs						
n	Random Draw		Linear Assignment ($\alpha = 0$)			
			over Ω		within each Ω_h	
20	-0.03	-0.08	-0.13	-0.11	-0.04	-0.23
50	-0.05	-0.09	-0.20	-0.15	-0.09	-0.25
100	-0.08	-0.12	-0.27	-0.22	-0.14	-0.33
200	-0.12	-0.16	-0.35	-0.30	-0.22	-0.40

Table 7

Relative design effects for stratified random sampling approaches with proportional allocation on the Voorst population. The method that performed best is shown in bold.

n	KMC Designs				
	Stratified GRTS	lcubestratified	Random Draw	Linear Assignment	
				over Ω	over Ω_h
20	0.96	0.82	0.83	0.71	0.75
50	0.80	0.70	0.75	0.55	0.63
100	0.74	0.60	0.59	0.48	0.51
200	0.62	0.49	0.52	0.37	0.42

linear assignment). The first method forces spatial spread over the population Ω and the second forces within stratum SB, with both approaches achieving desired stratum sample sizes. Results are presented in Table 6.

Stratified SRS produced SB averages close to zero for m_2 and $\bar{m}_h$. Better spatial spread was achieved using Stratified GRTS and stratified doubly balanced sampling, with their best results being within stratum SB. Random draw KMC achieved similar SB averages to stratified doubly balanced sampling, but the best KMC results were obtained using linear assignments. Both assignment strategies produced better m_2 and $\bar{m}_h$ averages than the other designs. Using (9) with $\alpha = 0$ achieved a high degree of spatial spread over Ω without compromising too much on average within stratum SB. Although not presented here, the linear assignment objective function (9) could be defined as a convex combination of m_2 and $\bar{m}_h$. This would allow us to jointly minimise overall SB and within stratum SB, producing SB averages between the limits in bold. In the following section, we consider the relationship between SB and the precision of an estimator.

5.2. Estimation

We consider estimating the total SOM concentrations in the Voorst study region using the unbiased Horvitz-Thompson estimator $\hat{\tau}$. To investigate precision, $\text{var}(\hat{\tau})$ was estimated using the empirical variance estimator (13). The design effect for estimating τ is defined as the variance of $\hat{\tau}$ under the proposed design divided by the variance of $\hat{\tau}$ under SRS. The design effect for stratified SRS with proportional allocation on the Voorst population is ≈ 0.75 . Hence, stratified sampling improves the precision of SRS. We report the relative design effect by dividing (13) by the variance of $\hat{\tau}$ under stratified SRS, where values less than one indicate greater precision in the proposed design. Results are presented in Table 7.

Other than stratified GRTS with $n = 20$, good gains in precision were made by the spatially balanced designs. The simple one point per cluster KMC design using random draws achieved similar results to stratified doubly balanced sampling. The KMC designs made the best gains with linear assignments. Slightly better results were obtained by forcing spread over the full population Ω , but achieving a high degree of spatial spread within each stratum was also an effective strategy for this population.

6. Conclusion

In this paper, we introduced a new one point per cluster sampling strategy that draws SB samples from both stratified and unstratified spatial populations. The design cannot achieve arbitrary inclusion probabilities but can draw equal probability samples and unequal probability samples via stratification. The proposed method was two-fold. First, the population was grouped into n

compact geographic clusters using the k -means algorithm. Then, one point was selected from each cluster using a linear assignment strategy that optimised the SB of the sample.

Numerical results on test populations with different spatial structures showed that the proposed method was competitive with several existing SB designs for achieving spatial spread and precise estimates of population totals. The technique also spread sample locations evenly over a stratified population, simultaneously achieving overall SB and within stratum SB. Our one point per cluster approach performed better than stratified SRS for estimating soil organic matter concentrations on the Voorst population, with relative design effects between 0.37 and 0.71. The proposed method also compared favourably with existing SB designs on this population.

The general framework of this two step approach is flexible because cluster definitions, the clustering algorithm and the linear assignment objective function are arbitrary. This paper focused on spatial balance, so k -means clusterings and objective functions based on popular SB measures were natural choices. Incorporating auxiliary variables into the design and using different dissimilarity measures, clustering methods, and linear assignment objective functions could achieve various sampling objectives. This presents an interesting research direction for further study.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.csda.2023.107888>.

References

- Benedetti, R., Piersimoni, F., Postiglione, P., 2015. Spatially balanced sampling: a review and a reappraisal. *Int. Stat. Rev.* 85, 439–454.
- Benedetti, R., Piersimoni, F., 2017a. A spatially balanced design with probability function proportional to within sample distance. *Biom. J.* 59, 1067–1084.
- Benedetti, R., Piersimoni, F., Postiglione, P., 2017b. Alternative and complementary approaches to spatially balanced samples. *Metron* 75, 249–264.
- Borgwardt, S., Brieden, A., Gritzmann, P., 2017. An LP-based k -means algorithm for balancing weighted point sets. *Cent. Eur. J. Oper. Res.* 263, 349–355.
- Brus, D.J., de Gruijter, J.J., Groenigen, J.W., 2007. Designing spatial coverage samples using the k -means clustering algorithm. In: Lagacherie, P., McBratney, A.B., Voltz, M. (Eds.), *Digital Soil Mapping. An Introductory Perspective*. Elsevier, pp. 183–192.
- Brus, D.J., 2022. *Spatial Sampling with R*. CRC Press, New York.
- van Dam-Bates, P., Gansell, O., Robertson, B.L., 2018. Using balanced acceptance sampling as a master sample for environmental surveys. *Methods Ecol. Evol.* 9, 1718–1726.
- Dickson, M.M., Tillé, Y., 2016. Ordered spatially balanced sampling by means of the traveling salesman problem. *Comput. Stat.* 31, 1359–1372.
- Duff, I.S., Koster, J., 2001. On algorithms for permuting large entries to the diagonal of a sparse matrix. *SIAM J. Matrix Anal. Appl.* 22, 973–996.
- Dumelle, M., Kincaid, T.M., Olsen, A.R., Weber, M.H., 2022. *spurvey: Spatial survey design and analysis*. R package version 5.3.0.
- Grafström, A., 2011. Spatially correlated Poisson sampling. *J. Stat. Plan. Inference* 142, 36–41.
- Grafström, A., Lundström, N.L.P., Schelin, L., 2012. Spatially balanced sampling through the pivotal method. *Biometrics* 68, 514–520.
- Grafström, A., Lundström, N.L.P., 2013a. Why well spread probability samples are balanced. *Open J. Stat.* 3, 36–41.
- Grafström, A., Tillé, Y., 2013b. Doubly balanced sampling with spreading and restitution of auxiliary totals. *Environmetrics* 14, 120–131.
- Grafström, A., Schelin, L., 2014. How to select representative samples. *Scand. J. Stat.* 41, 277–290.
- Grafström, A., Matei, A., 2018. Spatially balanced sampling of continuous resources. *Scand. J. Stat.* 45, 792–805.
- Grafström, A., Lisi, J., Prentius, W., 2022. *BalancedSampling: Balanced and Spatially Balanced Sampling*. R package version 1.6.3.
- Kermorvant, C., D'Amico, F., Bru, N., Caillil-Milly, N., Robertson, B., 2019. Spatially balanced sampling designs for environmental surveys. *Environ. Monit. Assess.* 191, 524.
- Larsen, D.P., Olsen, A.R., Stevens Jr., D.L., 2008. Using a master sample to integrate stream monitoring programs. *J. Agric. Biol. Environ. Stat.* 13, 243–254.
- Malinen, M.I., Fränti, P., 2014. Balanced k -means for clustering. In: Fränti, P., Brown, G., Loog, M., Escolano, F., Pelillo, M. (Eds.), *Structural, Syntactic, and Statistical Pattern Recognition. S+SSPR 2014*. In: *Lecture Notes in Computer Science*, vol. 8621. Springer, Berlin, Heidelberg.
- Neyman, J., Scott, E.L., 1958. Statistical approach to problems of cosmology. *J. R. Stat. Soc., Ser. B, Stat. Methodol.* 20, 1–43.
- Robertson, B.L., Brown, J.A., McDonald, T., Jaksons, P., 2013. BAS: balanced acceptance sampling of natural resources. *Biometrics* 3, 776–784.
- Robertson, B.L., McDonald, T., Price, C.J., Brown, J.A., 2017. A modification of balanced acceptance sampling. *Stat. Probab. Lett.* 129, 107–112.
- Robertson, B.L., McDonald, T., Price, C.J., Brown, J.A., 2018. Halton iterative partitioning: spatially balanced sampling via partitioning. *Environ. Ecol. Stat.* 25, 305–323.
- Robertson, B.L., van Dam-Bate, P., Gansell, O., 2021a. Halton iterative partitioning master frames. *Environ. Ecol. Stat.* 29, 55–76.
- Robertson, B.L., Reale, M., Price, C.J., Brown, J.A., 2021b. Quasi-random ranked set sampling. *Stat. Probab. Lett.* 171, 109029.
- Robertson, B.L., Ozturk, O., Kravchuk, O., Brown, J.A., 2022. Spatially balanced sampling with local ranking. *J. Agric. Biol. Environ. Stat.* 27, 622–639.
- Stevens Jr., D.L., Olsen, A.R., 1999. Spatially restricted surveys over time for aquatic resources. *J. Agric. Biol. Environ. Stat.* 4, 415–428.
- Stevens Jr., D.L., Olsen, A.R., 2003. Variance estimation for spatially balanced samples of environmental resources. *Environmetrics* 14, 593–610.
- Stevens Jr., D.L., Olsen, A.R., 2004. Spatially balanced sampling of natural resources. *J. Am. Stat. Assoc.* 99, 262–278.
- Theobald, D.M., Stevens, D.L., White Jr., D.E., Urquhart, N.S., Olsen, A.R., Norman, J.B., 2007. Using GIS to generate spatially balanced random survey designs for natural resource applications. *Environ. Manag.* 40, 134–146.
- Thompson, S.K., 2012. *Sampling*, third edition. Wiley, New York.
- Tillé, Y., Dickson, M.M., Espa, G., Giuliani, D., 2018. Measuring the spatial balance of a sample: a new measure based on the Moran's I index. *Spat. Stat.* 23, 182–192.
- Walvoort, D.J.J., Brus, D.J., de Gruijter, J.J., 2010. An R package for spatial coverage sampling and random sampling from compact geographical strata by k -means. *Comput. Geosci.* 36, 1261–1267.